

4

B

$$v^2 = u^2 + 2as, \quad 0 = u^2 + 2(-9.81)(12.7)$$

$$v = u + at, \quad 0 = \sqrt{2(9.81)(12.7)} + (-9.81)t$$

$$t = 1.61\text{s}, \quad \text{total time} = 2t = 2(1.61) = 3.22\text{ s}$$

5

C

$$\text{Gradient} = \text{acceleration} = (12.0 - 3.0)/15.0 = 9.0/15.0$$